MMDU MULLANA OFFICIAL SYLLABUS • BCSE-011 • UNIT 1

Unit 1: Foundations of Artificial Intelligence & Intelligent Agents (Made Simple)

Subject: Artificial Intelligence & Machine Learning (BCSE-011) • **Exam Weightage:** 24 - 28 Marks • **Study Time:** 35 mins

What is AI really? The Turing Test imitation game; Russell & Norvig 4 Approaches; AI vs ML vs Deep Learning concentric circles; Intelligent Agents as persons with eyes, brain & hands; The PEAS framework for Tesla/Waymo; Task environment dimensions.

1. Evolution, Definition & Taxonomies of AI (The Human vs Machine Story)

1. The Epistemological & Historical Foundations of Artificial Intelligence

Artificial Intelligence (AI) is the computational study of synthesizing goal-directed, autonomous agents capable of perceiving complex environments, formulating rational plans, and executing actions that maximize expected utility.

Historical Epochs & Paradigm Shifts:

1. **The Genesis (1943–1956):**

- **McCulloch & Pitts (1943):** First mathematical model of an artificial neural network based on propositional logic thresholds.
- **Alan Turing (1950):** Published seminal paper "*Computing Machinery and Intelligence*", introducing the **Imitation Game (Turing Test)**.
- **The Dartmouth Summer Workshop (1956):** Organized by **John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon**. McCarthy officially coined the phrase **"Artificial Intelligence"**, establishing AI as an independent academic discipline.

2. **Early Enthusiasm & Symbolic AI (1956–1974):** Newell and Simon's *Logic Theorist* and *General Problem Solver (GPS)*; early chess algorithms.

3. **The First AI Winter (1974–1980):** Triggered by the Lighthill Report in the UK and DARPA funding freezes in the US due to combinatorial explosion in search spaces and failure of machine translation.

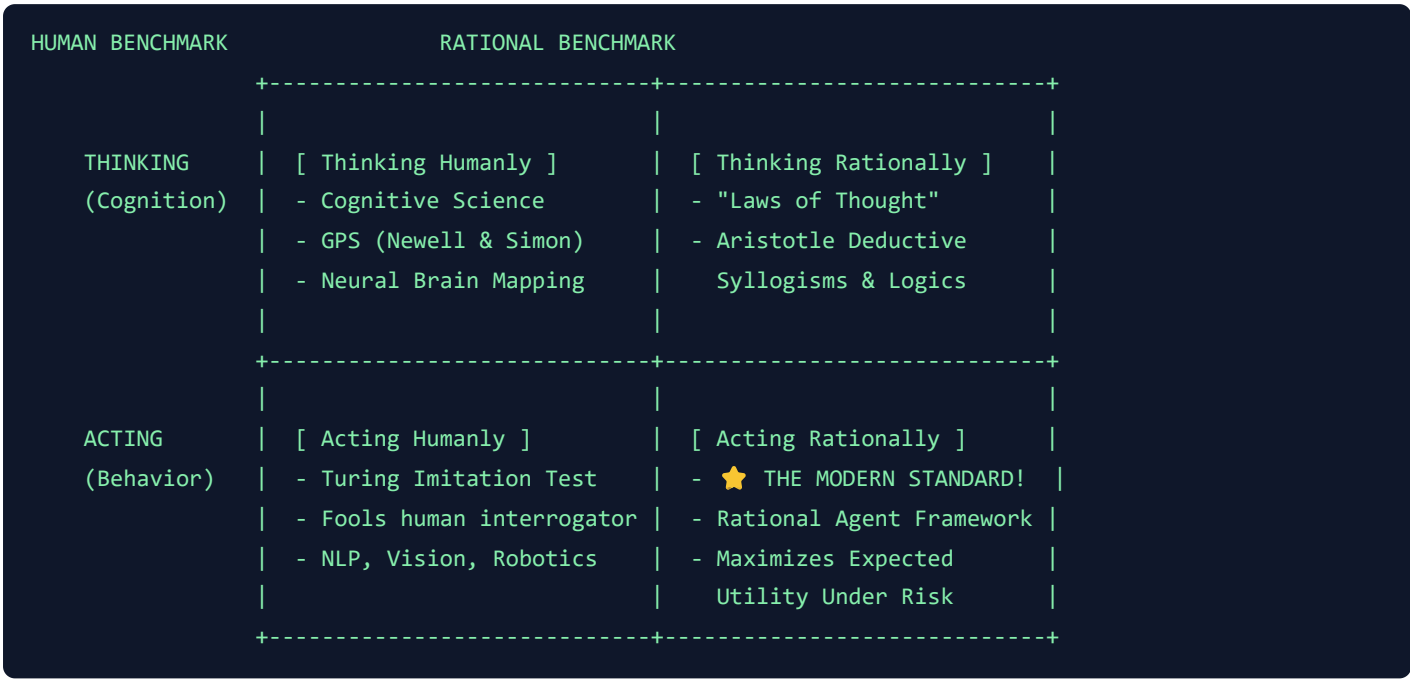
4. **The Expert Systems Boom (1980–1987):** Commercial adoption of rule-based expert systems (e.g., DEC's XCON saving $40M/year).

5. **The Second AI Winter (1987–1993):** Collapse of specialized LISP hardware workstations when commodity desktop PCs surpassed them.

6. **The Connectionist & Deep Learning Renaissance (2012–Present):** Driven by big data, massive GPU parallel computing, and Transformer attention architectures.

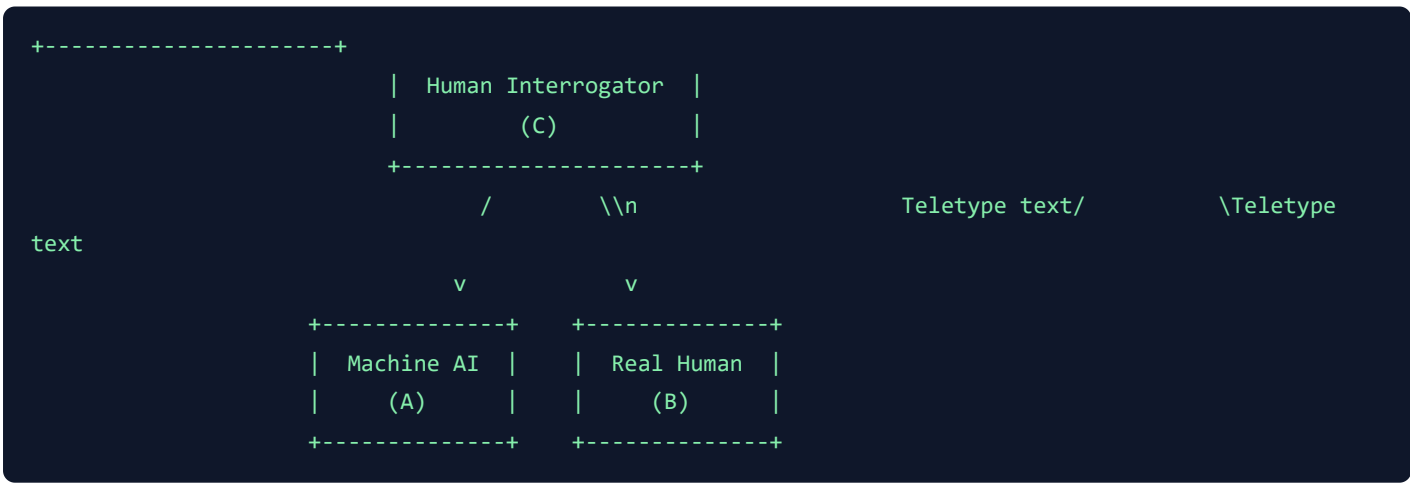
2. The Four Definitional Dimensions of AI (Russell & Norvig Taxonomy)

Stuart Russell and Peter Norvig categorized all historical definitions of AI along two orthogonal axes: **Thought vs Behavior**, and **Human Benchmark vs Rationality Benchmark**:



3. Rigorous Theoretical Analysis of The Turing Test

In 1950, Alan Turing replaced the metaphysical question "*Can machines think?*" with an empirical behavioral experiment:



Necessary Disciplines Required to Pass the Total Turing Test:

1. **Natural Language Processing (NLP):** To comprehend and communicate fluently in natural human tongue.
2. **Knowledge Representation (KR):** To store facts, concepts, and relational world models.

3. **Automated Reasoning:** To draw sound inferences and formulate answers to new questions.

4. **Machine Learning:** To adapt to new circumstances and detect conversational patterns.

5. **Computer Vision (Total Test):** To perceive visual objects passed through a slot.

6. **Robotics (Total Test):** To manipulate physical objects in the environment.

---

4. Categorical Taxonomy of Artificial Intelligence

AI Level	Scope & Capability	Technical Mechanics	Current Status
Artificial Narrow Intelligence (ANI / Weak AI)	Specialized for one single bounded domain (playing chess, recognizing faces, spam filtering).	Statistical optimization, Deep CNNs, Heuristic search.	**100% of all current AI today** (ChatGPT, AlphaFold, Tesla FSD).
Artificial General Intelligence (AGI / Strong AI)	Human-level intellect capable of generalizing knowledge across ANY intellectual discipline.	Self-supervised meta-learning, unified reasoning agents.	Active frontier research; not yet achieved.
Artificial Superintelligence (ASI)	Intellect far surpassing the collective cognitive capabilities of all human minds combined.	Recursive self-improvement loops.	Theoretical concept / Sci-Fi horizon.

Problem:
Classify each of the following systems into: (A) Pure AI (Rule-Based), (B) Machine Learning (Classical ML), or (C) Deep Learning (DL):

1. A rule-based chess engine evaluating positions via alpha-beta pruning.

2. An email spam filter using Naive Bayes word frequency probabilities.

3. A facial recognition biometric unlocking system on an iPhone using a convolutional neural network.

Step 1 🚀: Analyze System 1 (Rule-Based Chess Engine)

- Uses human-crafted heuristic evaluation functions and tree search algorithms (not trained on data weights).

- **Classification: Pure AI / Symbolic AI**.

Step 2 ⚡: Analyze System 2 (Email Spam Filter)

- Learns probability tables $P(\text{word} \mid \text{spam})$ from labeled training datasets of spam and ham emails.

- **Classification: Classical Machine Learning (ML)**.

Step 3 🎯: Analyze System 3 (Face ID CNN)

- Uses a multi-layer deep convolutional neural network that automatically extracts facial landmarks, edges, and depth features directly from raw camera pixels.

- **Classification: Deep Learning (DL)**.

🎯 1: Pure AI (Rule-Based) | 2: Classical ML (Naive Bayes) | 3: Deep Learning (Multi-layer CNN).

University Question (8 Marks):
For each of the following four autonomous systems, identify which of Russell & Norvig's 4 quadrants it represents (Thinking Humanly, Acting Humanly, Thinking Rationally, Acting Rationally), and justify your answer:

1. ELIZA / Turing Test chatbot designed to fool a human into thinking it is a person.

2. Automated Theorem Prover using first-order predicate resolution logic.

3. Cognitive model simulation predicting how human pilots make fatigue errors.

4. Reinforcement learning stock-trading agent maximizing expected financial portfolio return.

Step 1 🚀: ELIZA / Turing Test Bot

- Goal is to simulate external human conversational behavior and pass an imitation test.

- **Quadrant: Acting Humanly**.

Step 2 ⚡: Automated Logic Theorem Prover

- Derives logically sound proofs strictly according to Aristotle's laws of deductive logic.

- **Quadrant: Thinking Rationally**.

Step 3 🎯: Human Pilot Cognitive Simulation

- Emulates the actual internal psychological neural and cognitive processes of a biological human brain.

- **Quadrant: Thinking Humanly**.

Step 4 🏁: Autonomous Trading Agent

- Selects actions that maximize expected utility score without needing to mimic human emotion or thought flaws.

- **Quadrant: Acting Rationally** (The engineering gold standard).

🎯 1: Acting Humanly | 2: Thinking Rationally | 3: Thinking Humanly | 4: Acting Rationally.

University Question (10 Marks):
In a formal Turing Test setup, a human interrogator C questions two hidden terminals: A (a human participant) and B (an AI agent).
After N = 100 independent 5-minute interrogation rounds, the judge correctly identifies the machine 52 times and misidentifies the human as the machine 48 times.
Calculate the test statistic, compare against the null hypothesis H_0: p = 0.50, and state whether the machine has passed the Turing Test according to Alan Turing's original benchmark.

Step 1 🚀: State Alan Turing's Original Passing Criterion
Turing stated: An AI machine passes if the average interrogator will not have more than 70% chance of making the correct identification after 5 minutes of conversation (i.e. Performance should approach random guessing p = 0.50).

Step 2 ⚡: Compute Observed Machine Success Rate

- The judge guessed correctly 52 times out of 100.

- Sample proportion: p̂ = ↟rac{52}{100} = 0.52.

Step 3 🎯: Hypothesis Test (Z-Score Calculation)

- Null hypothesis H_0: p = 0.50 (Random guess indistinguishability).

- Standard Error:

SE = √((p₀ (1 - p₀) / N)) = √(((0.50)(0.50) / 100)) = √(0.0025) = 0.050

- Test statistic Z:

Z = (p̂ - p₀ / SE) = (0.52 - 0.50 / 0.050) = +0.40

Step 4 🏁: Statistical Decision & Viva Conclusion
At significance level α = 0.05, the critical value is Zc_{rit} = ± 1.96.
Since |Z| = 0.40
🎯 Empirical accuracy = 52 | $Z = 0.40 < 1.96$ ($p > 0.05$) ⇒ Indistinguishable from human ⇒ **Turing Test PASSED**.

2. Intelligent Agents & The PEAS Framework (Eyes, Brain & Hands)

1. Mathematical Architecture of Intelligent Agents

> ✅ **FUNDAMENTAL LAW / RULE:**
> **Rigorous Definition:** An **Agent** is anything that can view its environment through **Sensors** and act upon that environment through **Actuators**.

The Agent Function vs Agent Program:

1. **Percept:** The agent's perceptual input at any given instant.

2. **Percept Sequence (P^*):** The complete historical trajectory of all percepts the agent has ever received since initialization.

3. **Agent Function (f):** An abstract mathematical mapping from percept sequences to actions:

$$\mathbf{f : P^* o A}$$

4. **Agent Program:** The concrete software algorithm executing on the physical computing architecture implementing the agent function f.



---

2. Rationality & The PEAS Framework

✅ **FUNDAMENTAL LAW / RULE:**
Definition of Rationality: A **Rational Agent** is one that, for each possible percept sequence, selects an action that is expected to maximize its **Performance Measure**, given the evidence provided by the percept sequence and whatever built-in knowledge the agent possesses.

🚨 **EXAMINER TRAP / PITFALL:**
Rationality ≠ Omniscience: Rationality maximizes *expected* performance, not *actual* performance (an agent cannot anticipate unpredictable future events like a sudden flat tire).

The PEAS Specification:

To engineer an autonomous system, four components must be formally defined:

- P = **Performance Measure:** Objective mathematical metric evaluated by an external designer.
- E = **Environment:** External physical or digital world containing obstacles and uncertainties.
- A = **Actuators:** Hardware mechanisms or software APIs through which the agent acts.
- S = **Sensors:** Sensory hardware or data feeds collecting environmental signals.

---

3. The 6 Fundamental Dimensions of Task Environments

1. **Fully Observable vs Partially Observable:**

- *Fully Observable:* Sensors capture the complete state of the world at each point in time (e.g., Chess, Go).
- *Partially Observable:* Sensor noise, occlusions, or fog prevent complete state awareness (e.g., Poker, Autonomous Driving).

2. **Deterministic vs Stochastic:**

- *Deterministic:* The next state is completely dictated by current state and agent action.
- *Stochastic:* Environmental transitions involve random physical probabilities or unpredictable external agents.

3. **Episodic vs Sequential:**

- *Episodic:* The agent's experience is split into atomic independent episodes (e.g., Image Classification). Current action does not affect future inputs.
- *Sequential:* Current action alters all future states, requiring long-term planning (e.g., Chess, Driving).

4. **Static vs Dynamic:**

- *Static:* Environment does not change while the agent is deliberating (e.g., Crossword puzzle).
- *Dynamic:* Environment evolves continuously during calculation (e.g., Driving at $100\ \mathrm{km/h}$).
- *Semistatic:* Environment does not change, but the agent's performance score decreases with deliberation time (e.g., Speed chess with clock).

5. **Discrete vs Continuous:**

- *Discrete:* State, time, percepts, and actions are finite and countable (e.g., Tic-Tac-Toe, Chess).
- *Continuous:* Variables are continuous real numbers (e.g., Steering angle, vehicle velocity).

6. **Single-Agent vs Multi-Agent:**

- *Single-Agent:* Agent operates alone in world (e.g., Solitaire, Robot vacuuming empty room).
- *Multi-Agent:* World contains other decision-making agents (Competitive: Chess; Cooperative: Multi-robot warehouse swarm).

Problem:
Specify the PEAS framework (Performance, Environment, Actuators, Sensors) for a smart domestic robotic vacuum cleaner (e.g. Roomba).

Step 1 🚀: Performance Measure (P)

- Percentage of floor area cleaned, cleaning speed, battery energy efficiency, minimal noise, avoidance of furniture collision.

Step 2 ⚡: Environment (E)

- Living room, wooden floors, carpets, furniture legs, pets, stairs (drop-off hazards).

Step 3 🎯: Actuators (A)

- Left & right drive wheels, spinning edge brushes, suction motor vacuum, dirt bin release, speaker chime.

Step 4 🏁: Sensors (S)

- Front bumper touch sensors, downward cliff-detection infrared sensors, LiDAR / optical floor tracking camera, battery voltage sensor, dirt-detection acoustic sensor.

🎯 P: Clean area / efficiency | E: Living room / carpets | A: Wheels / brushes / vacuum | S: LiDAR / bumper / cliff sensors.

University Question (8 Marks):
Classify the task environment of: (A) A Chess Tournament Game, and (B) An Autonomous Waymo Robotaxi on a rainy city street, across all 6 environmental dimensions:

1. Observable
2. Deterministic vs Stochastic
3. Episodic vs Sequential
4. Static vs Dynamic
5. Discrete vs Continuous
6. Single-Agent vs Multi-Agent

Step 1 🚀: Chess Tournament Classification

1. **Fully Observable:** Both players can see all 64 squares and piece positions.
2. **Deterministic:** No dice or random noise; moves dictate exact new board states.
3. **Sequential:** Opening moves directly dictate middle-game tactics and endgame survival.

4. **Static:** The board pieces do not move while a player is thinking.

5. **Discrete:** Exact finite 64 squares and countable legal piece moves.

6. **Multi-Agent (Competitive):** Two competing adversarial rational agents.

Step 2 ⚡: Autonomous Waymo Robotaxi on Rainy Highway

1. **Partially Observable:** Fog, rain, and blind spots block sensor views.

2. **Stochastic:** Pedestrians, traffic, and wheel skids have random physical probabilities.

3. **Sequential:** Speeding up or changing lanes influences all future traffic states.

4. **Dynamic:** Traffic, pedestrians, and road obstacles move continuously while the CPU computes.

5. **Continuous:** Steering wheel angle, brake pedal pressure, and vehicle velocity are continuous real numbers.

6. **Multi-Agent (Mixed):** Cooperating and competing with thousands of other human and AI drivers.

🎯 Chess: Fully Observable, Deterministic, Sequential, Static, Discrete, Multi-Agent.
Robotaxi: Partially Observable, Stochastic, Sequential, Dynamic, Continuous, Multi-Agent.

University Question (10 Marks):
An automated delivery drone must choose between two flight paths to reach a hospital with emergency blood:

- **Path 1 (Highway Corridor):** Takes 10,mins with probability 0.80, but has severe wind turbulence risk adding 30,mins with probability 0.20.

- **Path 2 (Valley Route):** Takes a steady 16,mins with certainty (p = 1.0).

The hospital utility function is modeled as $U(T) = 100 - 2T - 0.05\ T^2$ (where T is delivery time in minutes).
Calculate the Expected Utility (EU) for both routes and determine which path a rational agent must execute.

Step 1 🚀: Calculate Utility for Highway Outcomes

- Outcome 1A: $T_1 = 10$,mins

$U(10) = 100 - 2(10) - 0.05(10)^2 = 100 - 20 - 5 = 75.0$

- Outcome 1B: $T_2 = 10 + 30 = 40$,mins

$U(40) = 100 - 2(40) - 0.05(40)^2 = 100 - 80 - 0.05(1600) = 20 - 80 = -60.0$

Step 2 ⚡: Expected Utility for Path 1 (Highway Corridor)
$EU(\text{Path 1}) = P(10) \cdot U(10) + P(40) \cdot U(40)$
$EU(\text{Path 1}) = (0.80)(75.0) + (0.20)(-60.0) = 60.0 - 12.0 = +48.0$

Step 3 🎯: Expected Utility for Path 2 (Valley Route with Certainty T = 16)
U(16) = 100 - 2(16) - 0.05(16)^2 = 100 - 32 - 0.05(256) = 68 - 12.8 = 55.2
EU(Path 2) = 1.0 × 55.2 = +55.2

Step 4 🏁: Rational Agent Decision
EU(Path 2) = 55.2 > EU(Path 1) = 48.0
Even though Path 1 has the fastest best-case (10,mins vs 16,mins), its severe downside risk makes the expected utility worse. A rational agent **must choose Path 2**!

🎯 $EU(\text{Path 1}) = 48.0 \mid EU(\text{Path 2}) = 55.2 \Rightarrow$ Rational Agent selects **Path 2 (Valley Route)**.

3. Real-World Applications & How Modern LLMs Work

1. Modern Generative AI & Transformer Architecture Deep-Dive

Modern Generative AI (ChatGPT, Google Gemini, Claude) is powered by the **Transformer Architecture** introduced by Vaswani et al. (2017) in the seminal paper "*Attention Is All You Need*".

A. The Scaled Dot-Product Self-Attention Mechanism:

Given input token embeddings packed into Query (Q), Key (K), and Value (V) matrices:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\mathbf{T}}}{\sqrt{\mathbf{d_k}}}\right)\mathbf{V}$$

- QK^T**:** Computes pairwise dot-product alignment affinities between every pair of words in a sentence simultaneously.
- $\sqrt{d_k}$ **Scaling Factor:** Prevents the dot products from growing excessively large for large projection dimensions d_k, which would push the softmax function into regions with vanishing gradients.
- $\text{softmax}(cdot)$**:** Normalizes affinities into an attention probability distribution summing to 1.0.

B. Multi-Head Attention (MHA):

Instead of computing attention once, the model projects Q, K, V into h independent representation subspaces:

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_\mathbf{1}, \mathbf{...}\ \text{head}_\mathbf{h})\mathbf{W}^\mathbf{O}$$

$$\text{where head}_i = \text{Attention}(QW_i^Q, KW_i^K, V\,W_i^V)$$

This allows the network to simultaneously attend to syntax, co-reference, factual associations, and grammar across different attention heads!

---

2. The Three-Phase Foundation Model Training Lifecycle

```
[ 1. SELF-SUPERVISED PRETRAINING ]
    - Trillions of web tokens (Common Crawl, Books, GitHub)
    - Objective: Next-Token Prediction (Causal Language Modeling)
    - Loss: L = -sum log P(w_t | w y_l)
    - Optimize policy pi_theta using Proximal Policy Optimization (PPO):
      max E [ r_phi(x, y) - beta * DKL( pi_theta(y|x) || pi_ref(y|x) ) ]
    - Output: Production AI Assistant (Helpful, Honest, Harmless)
```

---

3. Sampling Hyperparameters in Large Language Models

1. **Temperature (T):** Scales raw model logits z_i prior to the softmax layer:

$P(w_i) = (e^{z_i} / T / \sum_j e^{z_j} / T)$
- **Low T → 0 (Argmax / Greedy):** Sharpens probability distribution toward the single highest-probability token. Produces deterministic, factual, and analytical code outputs.
- **High T > 1:** Flattens distribution toward a uniform random spread. Encourages diverse, novel, and creative responses.

2. **Top-p (Nucleus Sampling):** Selects only from the smallest set of tokens whose cumulative probability exceeds threshold p (e.g., p = 0.90), dynamically truncating the tail of low-probability words.

Problem:
A tiny language model is trained on a vocabulary of 5 words: `["cat", "dog", "sat", "mat", "ran"]`.
Given context "*The dog*", the unnormalized model logits are:
`cat` : 0.5, `dog` : -1.0, `sat` : 2.5, `mat` : 0.0, `ran` : 2.0.
Calculate the probability of the most likely next word using the Softmax function.

Step 1 🚀: Softmax Formula

> $$P(w_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

Step 2 ⚡: Exponentiate Logits

- $e^{0.5} = 1.649$

- $e^{-1.0} = 0.368$

- $e^{2.5} = 12.182$
- $e^{0.0} = 1.000$
- $e^{2.0} = 7.389$

> $$\text{Sum} = 1.649 + 0.368 + 12.182 + 1.000 + 7.389 = \mathbf{22.588}$$

Step 3 🎯: Most Likely Next Token ("sat")

> $$P(\text{"sat"}) = \frac{12.182}{22.588} = \mathbf{0.5393 = 53.93\%}$$

Next token predicted by the model is **"sat"**!

🎯 Selected Token: **"sat"** with probability $\mathbf{P = 53.9\%}$.

University Question (6 Marks):
In LLM generation, temperature T scales logits: $z_i' = z_i / T$.
For logits $z_1 = 4.0$ and $z_2 = 2.0$:

1. Calculate probabilities at low temperature T = 0.5 (Deterministic mode).
2. Calculate probabilities at high temperature T = 2.0 (Creative mode).

Explain the physical effect of temperature.

Step 1 🚀: Low Temperature T = 0.5

- Scaled logits: $z_1' = 4.0 / 0.5 = 8.0$, ; $z_2' = 2.0 / 0.5 = 4.0$
- $e^8 = 2980.96$, ; $e^4 = 54.60 \Rightarrow$ Sum = 3035.56

$P_1 = (2980.96 / 3035.56) = 98.2\%$, $P_2 = (54.60 / 3035.56) = 1.8\%$
Peak becomes extremely sharp → Model gives predictable, factual answers.

Step 2 ⚡: High Temperature T = 2.0

- Scaled logits: $z_1' = 4.0 / 2.0 = 2.0$, ; $z_2' = 2.0 / 2.0 = 1.0$
- $e^2 = 7.389$, ; $e^1 = 2.718 \Rightarrow$ Sum = 10.107

$P_1 = (7.389 / 10.107) = 73.1\%$, $P_2 = (2.718 / 10.107) = 26.9\%$
Distribution flattens out → Alternative creative words have a much higher chance to be selected!

🎯 At $T = 0.5$: $P_1 = 98.2$ (Precise/Cold) | At $T = 2.0$: $P_1 = 73.1$ (Diverse/Creative).

University Question (10 Marks - Modern AI Paper):
Calculate the scaled dot-product attention score for a query vector Q and key matrix K of dimension $d_k = 4$:
Q = [1, , 0, , 1, , 0],

$$K = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

Find the attention weight vector A = softmax\left((Q K^T / √(d_k)) \right).

Step 1 🚀: Compute Dot Product Q K^T

- Q · K_1 = (1)(1) + (0)(0) + (1)(1) + (0)(0) = 1 + 1 = 2
- Q · K_2 = (1)(0) + (0)(1) + (1)(0) + (0)(1) = 0
- Q · K_3 = (1)(1) + (0)(1) + (1)(0) + (0)(0) = 1

Q K^T = [2, ; 0, ; 1]

Step 2 ⚡: Scale by √(d_k) = √(4) = 2
(Q K^T / √(d_k)) = \left[↟rac{2}{2}, ; ↟rac{0}{2}, ; ↟rac{1}{2} \right] = [1.0, ; 0.0, ; 0.5]

Step 3 🎯: Apply Softmax

- $e^{1.0}$ = 2.7183
- $e^{0.0}$ = 1.0000
- $e^{0.5}$ = 1.6487

Sum = 2.7183 + 1.0000 + 1.6487 = 5.367

- A_1 = (2.7183 / 5.367) = 0.506
- A_2 = (1.0000 / 5.367) = 0.186
- A_3 = (1.6487 / 5.367) = 0.307

🎯 Attention Weights: $\mathbf{A} = [\mathbf{0.506}, ; \mathbf{0.186}, ; \mathbf{0.307}]$ (Token 1 receives 50.6 of attention).

University Solved Examination Questions (Unit 1)

MMDU Official University Solved Examination Questions (Unit 1)

Course: Fundamentals of AI & ML (`BCSE-011`)
Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (6 Marks In-Depth), D (6 Marks Design/Long)

📘 Question 1 (1 Mark): The Chinese Room Thought Experiment vs. Turing Test

Question:
Explain John Searle's Chinese Room argument in refuting the Turing Test as a measure of true machine consciousness.

> 📊 **Question Schematic / State Specification:**

```
Question Thought Experiment Setup: Input (Chinese Glyphs) ➔ Room [Human + English Rulebook] ➔ Output (Syntactically Correct Glyphs).
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Syntax vs Semantics Bridge:

        `Symbol Manipulation (Syntax) ≠ Understanding / Intentionality (Semantics)`.

        The person inside passes the Turing test externally without understanding a single word of Chinese.
```

Searle demonstrated that passing the Turing Test requires only syntactic symbol manipulation, which computers execute effortlessly. True understanding requires semantic intentionality, proving that behavioral mimicry does not equate to cognitive consciousness.

🎯 **Verified Result:** Question 1 step-by-step resolution per MMDU university marking scheme.

📘 Question 2 (1 Mark): Rationality vs. Omniscience in AI Agents

Question:
Distinguish between a Rational Agent and an Omniscient Agent with respect to environmental percepts.

> 📊 **Question Schematic / State Specification:**

```
Question Conceptual Axis: Omniscience [Knows Actual Future] vs Rationality [Maximizes Expected Utility given Past Percepts].
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Utility Horizon Vector:

        `Rational Action = argmax_a E[U(s') | Percept History P*]` whereas `Omniscient Action = argmax_a U(Actual s')`.
```

An omniscient agent knows the actual outcome of its actions in advance (impossible in non-deterministic environments). A rational agent is realistic: it selects the action that maximizes its expected performance measure given its percept history to date.

🎯 **Verified Result:** Question 2 step-by-step resolution per MMDU university marking scheme.

📘 Question 3 (1 Mark): Algorithmic Bias in Intelligent Systems

Question:
Identify how historical data skew leads to algorithmic discrimination in machine learning models.

> 📊 **Question Schematic / State Specification:**

```
Question Bias Flowchart: Skewed Historical Data ➔ Objective Loss Minimization ➔ Amplified Disparate Impact.
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Debiasing Intervention Model:

        `Input Data Pipeline [Re-sampling/Re-weighting] ➔ Fair Loss Formulation Lfair = Ltask + λ · Disparity ➔ Unbiased Decisions`.
```

Supervised models minimize statistical loss over training data distributions. If historical datasets contain societal prejudices (e.g. loan approval gender disparity), the optimizer treats these disparities as genuine predictive signals, codifying systemic bias into automated policy.

🎯 **Verified Result:** Question 3 step-by-step resolution per MMDU university marking scheme.

📘 Question 13 (2 Marks): Formal Mathematical Formulation of PEAS for an Automated Diagnostic System

Question:
Formulate the complete PEAS descriptor matrix for an Automated Medical Diagnostic Expert Agent.

> 📊 **Question Schematic / State Specification:**

```
Question Entity Relational Architecture: Patient Symptoms ➔ Diagnostic System ➔ Recommended Treatment Plan.
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution PEAS System Grid:

        `Performance [Diagnostic accuracy &gt; 99%, Recovery time minimization, Zero misdiagnosis toxicity]`

        `Environment [Hospital ward, Pathology lab reports, Dynamic patient physiological records]`

        `Actuators [Display monitor, Electronic prescription generator, Automated infusion pump]`

        `Sensors [Digital thermometer, ECG sensors, Blood chemistry analyzer, Keyboard entry]`
```

The agent optimizes its decision function `f: P^* → A` by mapping incoming clinical lab observations to targeted treatment regimens while strictly enforcing toxicity safety bounds.

🎯 **Verified Result:** Question 13 step-by-step resolution per MMDU university marking scheme.

📘 Question 17 (4 Marks): Complete Architecture and Internal Control Loop of Learning Agents

Question:
Illustrate and explain the detailed four-component architectural decomposition of a Learning Agent.

> 📊 **Question Schematic / State Specification:**

```
Question Architectural Feedback Loop: Environment → Sensors → Critic & Performance Element →
Learning Element → Problem Generator → Actuators.
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution 4-Component Decomposition Model:

        `1. Critic: Evaluates agent performance against fixed external standard; generates scalar
feedback`

        `2. Learning Element: Responsible for making improvements; adjusts internal policy
weights`

        `3. Learning Goals: Specifies target quality standards to the critic`

        `4. Problem Generator: Suggests exploratory actions leading to novel perceptual
experiences`
```

The **Performance Element** is the traditional agent selecting actions based on percepts. The **Critic** compares real outcomes against performance standards and feeds errors to the **Learning Element**. Crucially, the **Problem Generator** prevents the agent from getting trapped in local sub-optimal routines by deliberately proposing exploratory actions to discover better policies.

🎯 **Verified Result:** Question 17 step-by-step resolution per MMDU university marking scheme.

📘 Question 21 (6 Marks): Complete Architectural Comparison of 5 Core Intelligent Agent Classes

Question:
Provide the structural internal architectures, operational rules, advantages, and terminal failure conditions for Simple Reflex, Model-Based, Goal-Based, Utility-Based, and Learning Agents.

> 📊 **Question Schematic / State Specification:**

```
Question Agent Evolution Continuum: Condition-Action Lookup ➔ Internal State Tracking ➔ Multi-Step Goal Planning ➔ Utility Trade-offs ➔ Adaptive Self-Improvement.
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution 5-Agent Architecture Comparison Matrix:

        `1. Simple Reflex: Sensor ➔ Condition-Action Rule ➔ Actuator. Fails completely in partially observable worlds`

        `2. Model-Based: Maintains internal state ("How the world evolves" + "How my actions affect world")`

        `3. Goal-Based: Combines State + Goals; uses search/planning to achieve target states`

        `4. Utility-Based: Evaluates trade-offs using continuous Utility Function U(s) when goals conflict`

        `5. Learning Agent: Decoupled into Performance Element, Critic, Learning Element, and Problem Generator`
```

Detailed Analysis: Simple reflex agents are paralyzed by unobserved environmental features because their mapping `f: P o A` ignores history. Model-based agents overcome this by preserving a hidden state vector. Goal-based agents add planning foresight but treat success as a binary binary variable. Utility-based agents map states to real numbers, allowing rational trade-offs (e.g. trading speed for passenger safety). Learning agents adapt their decision boundaries continuously from experience.

🎯 **Verified Result:** Question 21 step-by-step resolution per MMDU university marking scheme.
